SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0611 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO5: *Introduction – NumPy Arrays – NumPy Basics- Math – Random – Indexing – Filtering – Statistics – Aggregation – saving data – Data analysis with Pandas – DataFrame – Combining – Indexing – File I/O – Grouping – Filtering – Sorting – Plotting (BL 3)*

Assessment Tool Weightage: 20%

QUESTIONS:4 **Total Marks:100**

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

I K.ASWINI(192312397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. Design a pseudocode algorithm to perform **data cleaning and transformation on time-series data**. The algorithm should handle missing timestamps, interpolate missing values, remove outliers, and resample data at fixed intervals. Include steps for indexing by time, filtering invalid entries, and ensuring chronological consistency while maintaining efficiency.

2. Write a detailed pseudocode algorithm to implement **feature selection** on a dataset. The algorithm should compute correlation between features, remove redundant or irrelevant attributes, and retain significant features. Include steps for normalization, threshold-based filtering, and validation to ensure improved model performance and reduced dimensionality.

3. Create a pseudocode algorithm to detect and handle **outliers in a dataset** using statistical methods such as Z-score or IQR. The algorithm should identify extreme values, decide whether to remove or cap them, and ensure that the resulting dataset maintains consistency and reliability for further analysis.

4. Write a pseudocode algorithm to implement **dimensionality reduction using Principal Component Analysis (PCA)**. The algorithm should include steps for standardizing the dataset, computing covariance matrix, extracting eigenvalues and eigenvectors, and selecting principal components. Ensure efficient matrix operations and validation of reduced dimensions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

1.Data Cleaning & Transformation for Time-Series Data

Pseudocode

ALGORITHM TimeSeries_Cleaning(Data)

INPUT: Raw dataset Data(time, value)
OUTPUT: Cleaned and resampled dataset CleanData

BEGIN
1. Convert 'time' column to datetime format
2. Remove duplicate timestamps

3. Sort Data by time (chronological order)

4. Set 'time' as index

5. HANDLE MISSING TIMESTAMPS
   a. Create complete time range from min(time) to max(time)
   b. Reindex dataset with full time range
   c. Missing timestamps → assign NULL values

6. HANDLE MISSING VALUES
   a. Apply interpolation (linear / spline)
   b. If still missing → forward fill / backward fill

7. REMOVE INVALID ENTRIES
   a. Filter values outside valid range (domain constraints)

8. OUTLIER DETECTION (IQR method)
   a. Q1 = 25th percentile, Q3 = 75th percentile
   b. IQR = Q3 - Q1
   c. LowerBound = Q1 - 1.5 * IQR
      UpperBound = Q3 + 1.5 * IQR
   d. Remove or cap values outside bounds

9. RESAMPLING
   a. Choose fixed interval (e.g., 1 minute / 1 hour)
   b. Aggregate using mean/sum:
      CleanData = Resample(Data, interval, method="mean")

10. Ensure no missing values remain

11. Return CleanData

END

Key Points

- Ensures chronological consistency
- Handles both missing timestamps & values
- Uses efficient vectorized operations

2. Feature Selection Algorithm

Pseudocode

ALGORITHM Feature_Selection(Data, Target, threshold_corr)

INPUT: Dataset Data(features, target)
OUTPUT: Reduced feature set SelectedFeatures

BEGIN

1. Separate features X and target Y

2. NORMALIZATION
   For each feature f in X:
      f = (f - mean(f)) / std(f)

3. COMPUTE CORRELATION MATRIX
   CorrMatrix = Correlation(X)

4. REMOVE REDUNDANT FEATURES

```
    For each pair (fi, fj) in CorrMatrix:
       If |corr(fi, fj)| > threshold_corr:
          Remove one feature (lower importance)

 5. REMOVE IRRELEVANT FEATURES
    For each feature f in X:
       Compute correlation with target: corr(f, Y)
       If |corr(f, Y)| < threshold_corr:
          Remove feature f

 6. OPTIONAL: FEATURE IMPORTANCE
    Train simple model (e.g., decision tree)
    Rank features based on importance
    Keep top-k features

 7. VALIDATION
    Train model using selected features
    Evaluate performance (accuracy / RMSE)

    If performance improves or stable:
       Accept SelectedFeatures
    Else:
       Adjust threshold and repeat

 8. Return SelectedFeatures
END
```

Key Points

- Reduces dimensionality
- Removes redundant + irrelevant features
- Includes validation step

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3. Outlier Detection & Handling

Pseudocode

```
ALGORITHM Outlier_Detection(Data, method)

INPUT: Dataset Data
OUTPUT: Clean dataset CleanData

BEGIN
   IF method == "Z-score":
      1. Compute mean μ and standard deviation σ
      2. For each value x in Data:
           Z = (x - μ) / σ
           If |Z| > 3:
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            Mark as outlier

    ELSE IF method == "IQR":
        3. Compute Q1 and Q3
        4. IQR = Q3 - Q1
        5. LowerBound = Q1 - 1.5 * IQR
           UpperBound = Q3 + 1.5 * IQR

        6. For each value x:
              If x < LowerBound OR x > UpperBound:
                  Mark as outlier

    7. HANDLE OUTLIERS
        For each outlier:
            OPTION 1: Remove value
            OPTION 2: Cap value:
                If x < LowerBound → x = LowerBound
                If x > UpperBound → x = UpperBound

    8. Return CleanData
END
```

Key Points

- Supports Z-score and IQR
- Allows removal or capping
- Maintains data consistency

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4. PCA (Principal Component Analysis) Algorithm

Pseudocode

```
ALGORITHM PCA_Reduction(Data, k)

INPUT: Dataset Data(n_samples, n_features), number of components k
OUTPUT: Reduced dataset Z

BEGIN
    1. STANDARDIZATION
        For each feature:
            X = (X - mean) / std

    2. COMPUTE COVARIANCE MATRIX
        CovMatrix = (X^T * X) / (n - 1)

    3. COMPUTE EIGENVALUES & EIGENVECTORS
        [EigenValues, EigenVectors] = EigenDecomposition(CovMatrix)
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    4. SORT COMPONENTS
       Sort eigenvalues in descending order
       Arrange eigenvectors accordingly

    5. SELECT TOP k COMPONENTS
       W = first k eigenvectors

    6. PROJECT DATA
       Z = X * W   (Reduced representation)

    7. VALIDATION
       Compute explained variance:
          ExplainedVariance = sum(top k eigenvalues) / total eigenvalues

       If ExplainedVariance >= threshold (e.g., 95%):
          Accept reduction
       Else:
          Increase k

    8. Return Z
END
```

Key Points

- Uses matrix operations (efficient)
- Preserves maximum variance
- Includes explained variance validation